Chapter 1 solutions

DM01-E01

There are $26 - 2 = 24$ usable letters, chosen distinctly in $24 \cdot 23 \cdot 22$ ways. The first digit has nine choices. The second has nine remaining digits, hence

$$24 \cdot 23 \cdot 22 \cdot 9 \cdot 9 = 983\,664.$$

DM01-E02

Choose the chair from eight, the vice chair from seven, then three members from the remaining six:

$$8 \cdot 7 \binom{6}{3} = 56 \cdot 20 = 1120.$$

Alternatively choose five people and assign the two ordered offices, giving $\binom{8}{5} \cdot 5 \cdot 4 = 1120$. Both descriptions retain ordered offices and unordered ordinary membership.

DM01-E03

BANANARAMA has ten letters, including five A's, two N's, and one each of B, R, M. Fix A at each end. The middle eight positions contain three A's, two N's, and B, R, M, giving

$$\frac{8!}{3!2!} = 3360.$$

Lin Cing https://github.com/xinhi6716/computer-science-self-study

DM01-E04

The right side chooses a representative from n people, then an arbitrary subset of the other $n - 1$, giving $n2^{n-1}$. The left side first chooses a size-k subset, then its representative, and sums over $k = 0, \ldots, n$. Each nonempty subset with a representative is counted exactly once by both methods, proving equality.

DM01-E05

Put $y_i = x_i - 2 \geq 0$. Then

$$y_1 + y_2 + y_3 + y_4 = 5.$$

Stars and bars gives

$$\binom{5 + 4 - 1}{4 - 1} = \binom{8}{3} = 56.$$

DM01-E06

$$C_6 = \frac{1}{7}\binom{12}{6} = \frac{924}{7} = 132.$$

For the exact reflection, swap the prefix through the first height -1 of an inadmissible balanced word. The result has seven openings and five closings. Conversely, in any such word, swap through its first height $+1$, recovering an inadmissible balanced word. These maps are inverse, so $\binom{12}{5} = 792$ words are inadmissible, leaving $924 - 792 = 132$.

Division by two has no equal-pairing justification. The actual class sizes are 132 and 792. Swapping all parentheses sends a nonempty admissible word to an inadmissible one, but many inadmissible words remain inadmissible, such as `())(()` and `)(())(`. A pair need not contain exactly one admissible word.

DM01-E07

Remove A's, leaving B, N, N, R, M with $5!/2! = 60$ orders. Each has six gaps; five A's choose distinct gaps:

$$60\binom{6}{5} = 360.$$

Deleting A's recovers the non-A order, and their original positions identify the selected gaps uniquely. Each resulting word is counted once.

With A at both ends, both exterior gaps are selected. Choose three of the four interior gaps:

$$60\binom{4}{3} = 240.$$

A's are indistinguishable, so do not multiply by ways to choose which two occupy the ends.

DM01-E08

Four runs have two 1-runs and two 0-runs, with either starting symbol. Compositions of six ones into two positive lengths number $\binom{5}{1} = 5$, and those of three zeros number $\binom{2}{1} = 2$. Thus

$$2 \cdot 5 \cdot 2 = 20.$$

Five runs beginning with 1 have three 1-runs and two 0-runs, giving $\binom{5}{2}\binom{2}{1} = 20$. Beginning with 0 gives two 1-runs and three 0-runs, giving $\binom{5}{1}\binom{2}{2} = 5$. The disjoint cases total 25.

Without zeros, only 111111 exists, with one run. The formula assumes both symbols occur, so substituting $\binom{-1}{k}$ is invalid. If neither symbol occurs, the empty word has zero runs.

DM01-E09

For nonempty shelves, permute seven books and choose two cuts among six gaps:

$$7!\binom{6}{2} = 5040 \cdot 15 = 75\,600.$$

Concatenating shelves in label order uniquely recovers the order and cuts.

Allowing empty shelves, choose $a + b + c = 7$, $a, b, c \geq 0$, in $\binom{9}{2} = 36$ ways, then cut a permutation to those lengths:

$$7!\binom{9}{2} = 181\,440.$$

Recording destinations only gives $3^7 = 2187$. Erasing internal order has fiber size $a!b!c!$, which varies with shelf lengths. No fixed divisor converts the previous counts into this one.

DM01-E10

Two of seven factors supply x, three supply $-2y$, and two supply z. The positional assignment count is

$$\frac{7!}{2!3!2!} = 210.$$

Each contributes weight $(-2)^3 = -8$, so the coefficient is $210(-8) = -1680$.

Set $x = y = z = 1$ to sum all coefficients, obtaining $(1 - 2 + 1)^7 = 0$. No set has negative size here. The positional choices have nonnegative counts; their signed weights are then added.

DM01-E11

All words number $\binom{11}{4} = 330$. Reflecting the prefix through the first height -1 changes one excess negative step to positive. The target has eight positives and three negatives, giving $\binom{11}{3} = 165$.

Every target word ends at five, so it first reaches $+1$. Reflect that prefix to recover a word with seven positives and four negatives whose first negative height is -1. The operations are inverse. The admissible count is

$$330 - 165 = 165.$$

The answer happens to be half for these parameters. This does not justify general division by two; with seven positives and seven negatives, the admissible proportion is $1/8$.

DM01-E12

A full binary tree with five internal nodes has six leaves. Encode a leaf by the empty word. If the left and right subtrees encode to U, V, encode their parent as $(U)V$. To invert, find the first closing parenthesis matching the initial opening parenthesis; the enclosed word is uniquely U and the remaining suffix is V. Recursively decode both subtrees. These operations are inverse, giving a bijection with five-pair admissible words and therefore

$$C_5 = \frac{1}{6}\binom{10}{5} = 42.$$

For a heptagon, $n + 2 = 7$, hence $n = 5$ and $C_5 = 42$, through the root-triangle recurrence. Six fixed-order factors need five binary combinations, also giving 42.

Seven factors need six internal nodes and give

$$C_6 = \frac{1}{7}\binom{12}{6} = 132.$$

Parameters must follow the structures rather than merely matching numerals in the questions.

DM01-E13

The map

$$(k, j, i) \longmapsto (k, j, i + 1)$$

gives $1 \le k < j < i + 1 \le n + 1$, a three-element subset of $\{1, \dots, n + 1\}$. Subtracting one from the third coordinate inverts it, recovering $k < j \le i$. Thus the count is

$$\binom{n+1}{3}.$$

At $n = 8$, this is $\binom{9}{3} = 84$. The lecture allows $k = j$ as well, giving 120. Their difference, 36, is exactly the number of pairs $1 \le j \le i \le 8$, namely $\binom{9}{2}$. For the stated integer domain $0 \le n < 2$, there are no triples and the out-of-range binomial convention gives zero.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

DM01-E14

Set $y_1 = x_1 - 2, y_2 = x_2 - 1, y_3 = x_3, y_4 = x_4$. Their nonnegative total is $11 - 2 - 1 = 8$. Ignoring the upper bound gives

$$\binom{8 + 4 - 1}{4 - 1} = \binom{11}{3} = 165.$$

For inadmissible solutions $y_4 \geq 4$, put $z_4 = y_4 - 4 \geq 0$. Total four gives $\binom{7}{3} = 35$. The answer is

$$165 - 35 = 130.$$

Each inadmissible solution uniquely corresponds to a total-four solution. Only one violation class is subtracted, so overlapping complements are not an issue.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 1.

Original educational synthesis by Lin Cing, licensed under CC BY 4.0.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study